

Joseph Lavond

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EDUCATION

University of North Carolina

Ph.D. Statistics and Operations Research

M.S. Statistics and Data Science

Chapel Hill, NC

Aug. 2020 – May 2025

Aug. 2020 – May 2024

California Polytechnic State University

B.S. Statistics

San Luis Obispo

Sep. 2016 – Mar. 2020

EXPERIENCE

Data Scientist

June 2025 – Present

Red Ventures

Fort Mills, SC

- Built and productionized ReplayOPE, an offline policy evaluation platform for contextual bandits enabling policy evaluation, automated recommendations, and experimentation for production decision systems
- Owned the end-to-end ML product lifecycle, including system architecture, Databricks deployment, CI/CD (GitHub Actions, Terraform), and development of a FastAPI-based service layer supporting programmatic access for LLM-driven workflows and future agent capabilities
- Developed advanced experimentation methodology including combinatorially purged cross-validation, warm starts, streaming evaluation, and statistically rigorous policy comparison
- Improved core offline RL infrastructure by fixing a 3+ year missing-values inference bug, implementing recency-based sample weighting, and modularizing data pipelines for robustness and scalability
- Productionized advanced ML methods including CatBoost and LightGBM enhancements (ordinal encoding), enabling improved performance on high-cardinality decisioning problems
- Drove cross-team adoption through onboarding, documentation, and automated publishing of searchable docs (MkDocs, GitHub Pages, Confluence), while contributing to LLM enrichment pipelines and internal analytics tooling

Graduate Data Science Intern

June 2024 – Aug. 2024

Red Ventures

Fort Mills, SC

- Implemented offline model-based Q-Learning, a reinforcement learning algorithm, demonstrating a typical 5% increase in revenue per session, translating to an estimated \$5M in additional business revenue
- Developed and integrated a simulation environment focused on Markov Decision Processes for website traffic
- Established a streamlined pipeline leveraging historical production data to generate simulations, resulting in a substantial increase in both the quantity and complexity of simulations, leading to robust solutions

Graduate Research Intern

June 2023 – Aug. 2023

Cisco

San Jose, CA

- Evaluated changes to prevent the occurrence of mode collapse in Generative Adversarial Networks, where samples are not diverse and did not learn to cover the training distribution
- Developed a more efficient way to train generative models with differential privacy at scale, enabling statistical guarantees for the protection of private training data

Graduate Info Technology Intern

June 2022 – Aug. 2022

Elevance Health, Inc. (Formerly Anthem)

Indianapolis, IN

- Implemented custom PyTorch semi-supervised Bayesian anomaly detection approach on Amazon Web Service (AWS) to identify exaggerated claims by providers for services offered
- Created modeling data set pipeline for 60M row database using PySpark SQL in Jupyter on Kubeflow

PUBLICATIONS

- Joseph Lavond. *Advancing Model Security, Data Privacy, and Performance for Widespread Adoption of Trustworthy Artificial Intelligence*. PhD thesis, University of North Carolina at Chapel Hill, 2025. Accepted
- Joseph Lavond, Minhao Cheng, and Yao Li. Feddecay: Balancing model performance and rapid personalization in federated learning with learning rate scheduling. 2025. Under Review
- Joseph Lavond, Minhao Cheng, and Yao Li. Trusted aggregation (TAG): Backdoor defense in federated learning. *Transactions on Machine Learning Research*, 2024

SKILLS

Machine Learning & Decisioning: Contextual Bandits, Offline Policy Evaluation (OPE), Reinforcement Learning, LightGBM, CatBoost, Ensemble Methods, Bayesian Inference

Programming & Data: Python, SQL, Pandas, NumPy, PySpark, SciPy

ML Systems & Infrastructure: FastAPI, Databricks, CI/CD (GitHub Actions), Terraform, Model Deployment, Experimentation Platforms

Data Engineering & Analytics: Feature Engineering, Data Pipelines, Statistical Testing, A/B Testing, Clustering, Segmentation